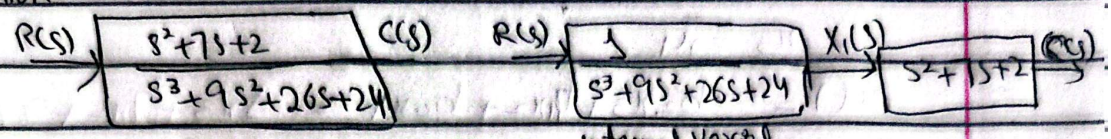
26-10-2026 Wednesday.

Name Eisha Gul
Reg.no. SP24-ELC-022
Subject Control systems.

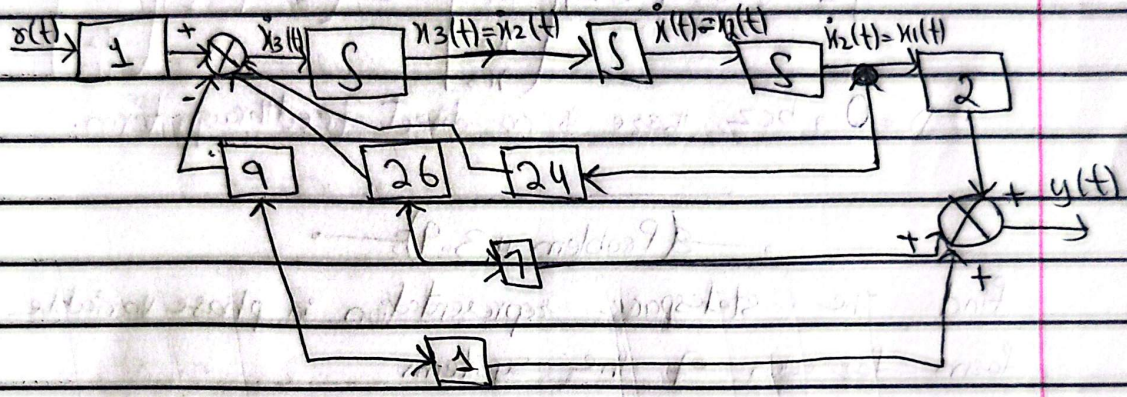
Assignment #02

(Example 3.5)

Find the state space representation of the transfer function.



(a)



$$G(S)=\frac{C_S}{R_S}=\frac{s^2+7s+2}{s^3+9s^2+26s+24}$$

$$\frac{X_1(S)}{R(S)}=\frac{1}{s^3+9s^2+26s+24}, \quad \frac{C_S}{X_1(S)}=s^2+7s+2$$

Find the state equations for the denominator block.

$X_1(S)=Y(S)$

$\dot{x}_1 = x_2$

$\dot{x}_2 = x_3$

$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + r$

$\dot{x} = Ax + Br$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

C(s) to $X_1(s)$ using numerator coefficients.

$b_0 = 2,\ b_1 = 7,\ b_2 = 1$

$C(s) = (b_2 s^2 + b_1 s + b_0) X_1(s) = (s^2 + 7s + 2) X_1(s)$

$c(t) = \ddot{x} + 7\dot{x} + 2x_1$

$x_1 = x_1$

$x_2 = \dot{x}_1$

$x_3 = \ddot{x}_1$

$c(t) = b_2 x_3 + b_1 x_2 + b_0 x_1 = 1x_3 + 7x_2 + 2x_1$

final output equation $\Rightarrow y = Cx + Dr$

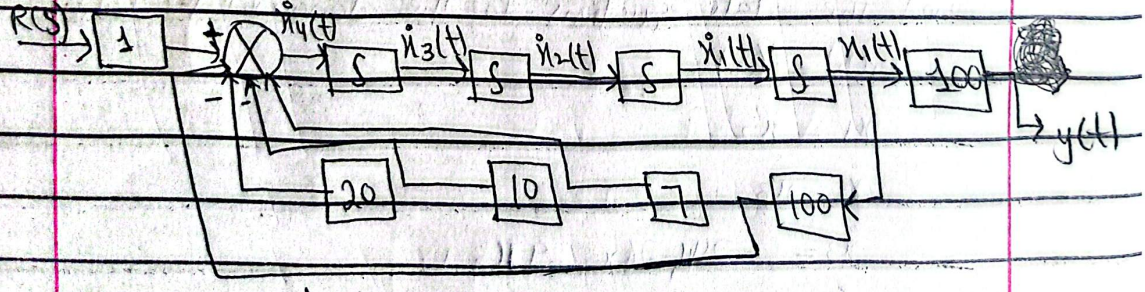
$$y = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$D = 0$, bcz there is no direct feedthrough term.

Problem #3.9

Find the statespace representation in phase variable form for each of the systems

(a) $R(s) \rightarrow \boxed{\dfrac{100}{s^4 + 20s^3 + 10s^2 + 7s + 100}} \rightarrow C(s)$

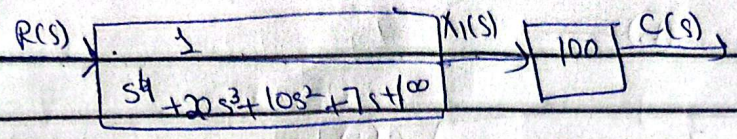


The denominator is of order $n = 4$

identify denominator coefficients.

$s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0 = s^4 + 20s^3 + 10s^2 + 7s + 100$

$a_3 = 20,\ a_2 = 10,\ a_1 = 7,\ a_0 = 100$



$$\frac{X_1(s)}{R(s)} = \frac{1}{s^4+20s^3+10s^2+7s+100}$$

$$\frac{C(s)}{X_1(s)} = 100 \Rightarrow C(s) = 100X_1(s)$$

find the state equations.

$$x_1 = x_1$$
$$\dot{x}_1 = x_2$$
$$\ddot{x}_1 = \dot{x}_2 = x_3$$
$$\dddot{x}_1 = \dot{x}_3 = x_4$$
$$\dot{x}_4 = -100x_1 - 7x_2 - 10x_3 - 20x_4 + r$$
$$\dot{x} = Ax + Br$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -100 & -7 & -10 & -20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} r$$

account for numerators

C(s) to $X_1(s)$

$$C(s) = 100X_1(s)$$

$$y(t) = c(t) = 100x_1$$

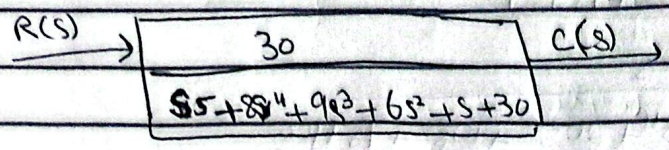
final output equation.

$$y = Cx + Dr$$

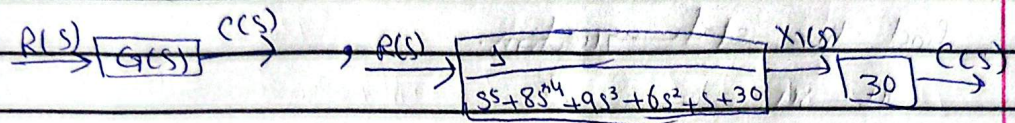
$$y = \begin{bmatrix} 100 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

(with $D = [0]$). Hint -

(b)



$$G(S) = \frac{C(S)}{R(S)} = \frac{30}{S^5+8S^4+9S^3+6S^2+S+30}$$



$$G(S) = \frac{X_1(S)}{R(S)} \cdot \frac{C(S)}{X_1(S)}$$

$$\frac{X_1(S)}{R(S)} = \frac{1}{S^5+8S^4+9S^3+6S^2+S+30}$$

$$\frac{C(S)}{X_1(S)} = 30$$

find the state equation

$x_1 = x_1$, $\dot{x}_1 = x_2$, $\dot{x}_2 = x_3$, $\dot{x}_3 = x_4$, $\dot{x}_4 = x_5$

$\dot{x}_5 = -30x_1 - x_2 - 6x_3 - 9x_4 - 8x_5 + r$

canonical form $\dot{x} = Ax + Br$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -30 & -1 & -6 & -9 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} r$$

account the numerator $y(t) = C(t) = 30x_1$

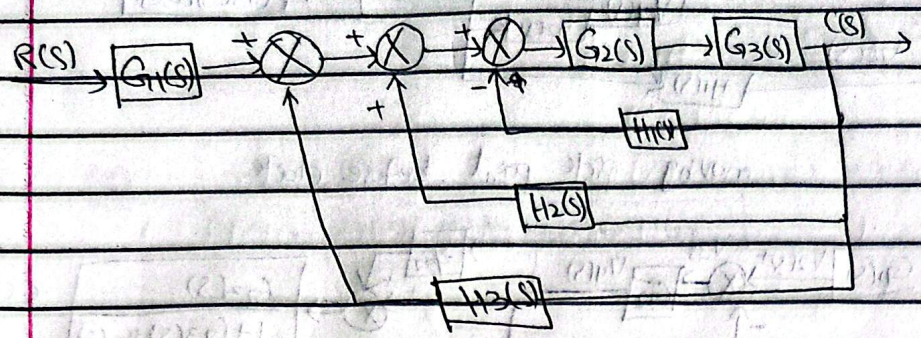
final output equation.

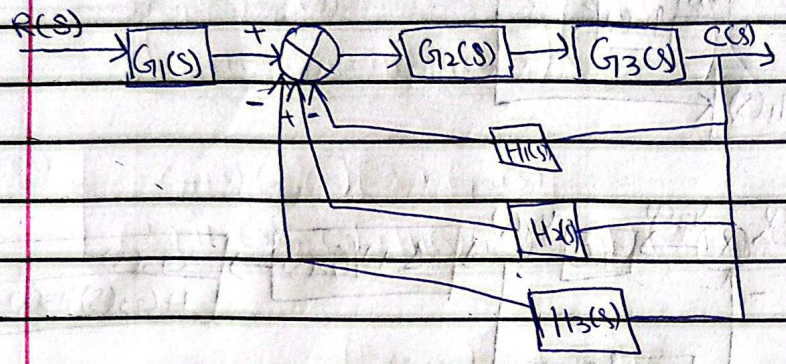
$y = Cx + Dr$

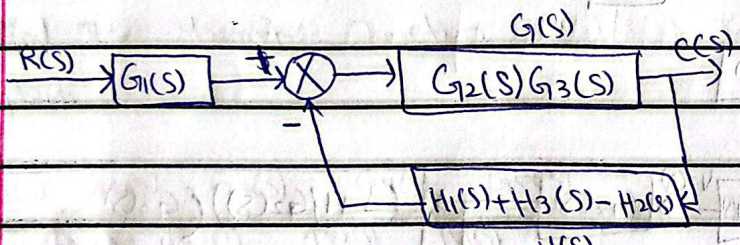
$$y = \begin{bmatrix} 30 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

Example # 5.1

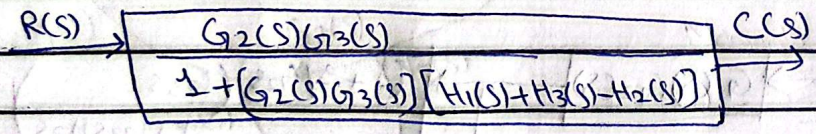
Block diagram reduction via familiar forms.



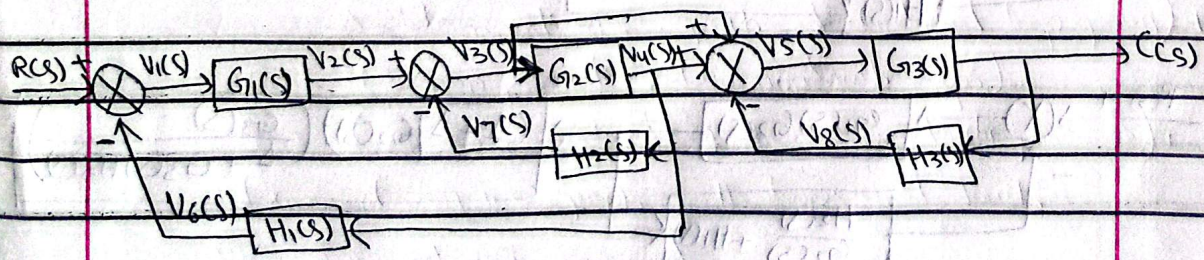


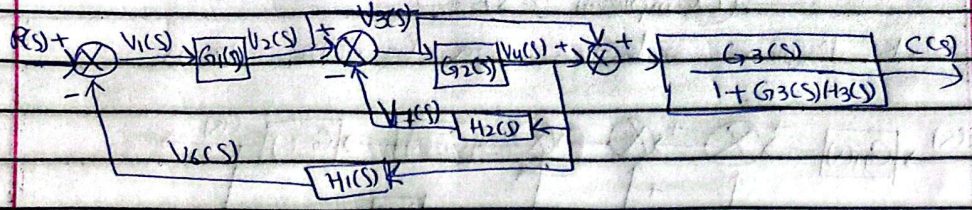


$$\frac{R(S)}{} = \frac{G(S)}{1 + G(S)H(S)}$$

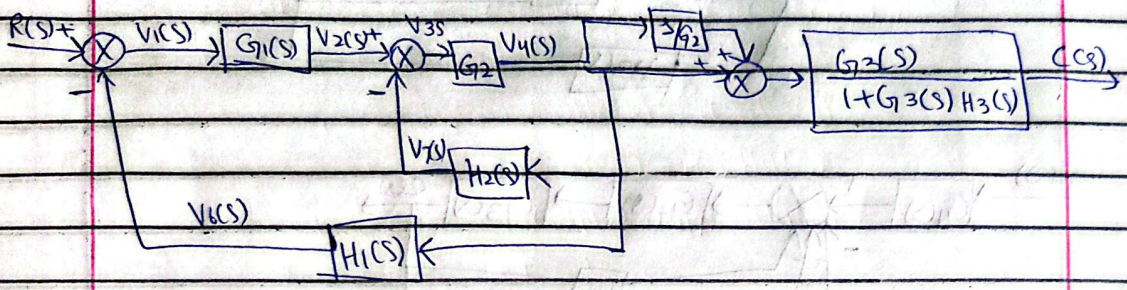


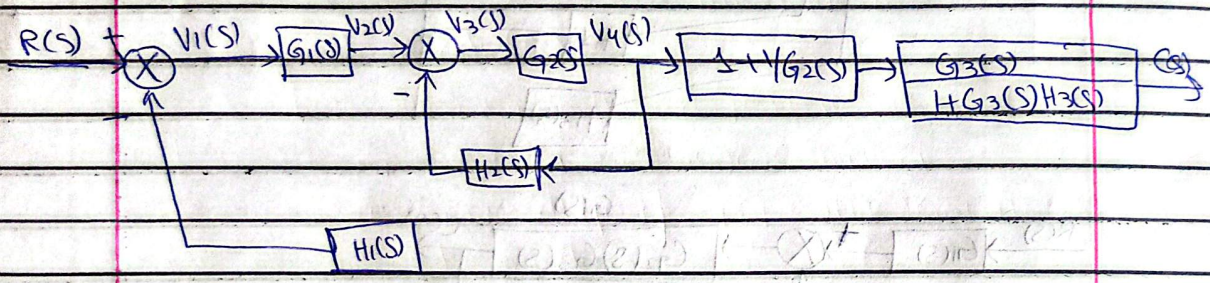
Example # 5.2

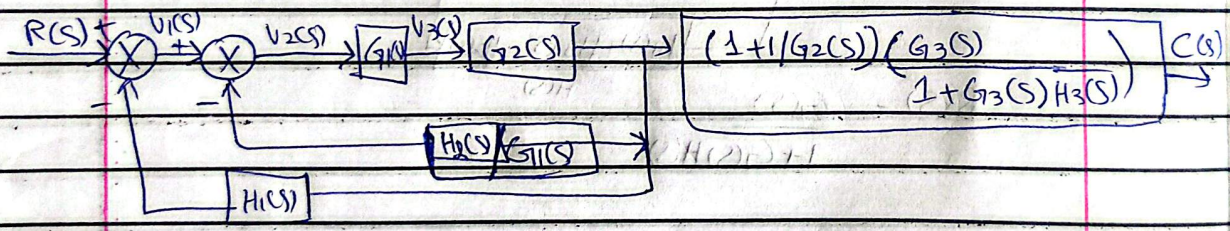


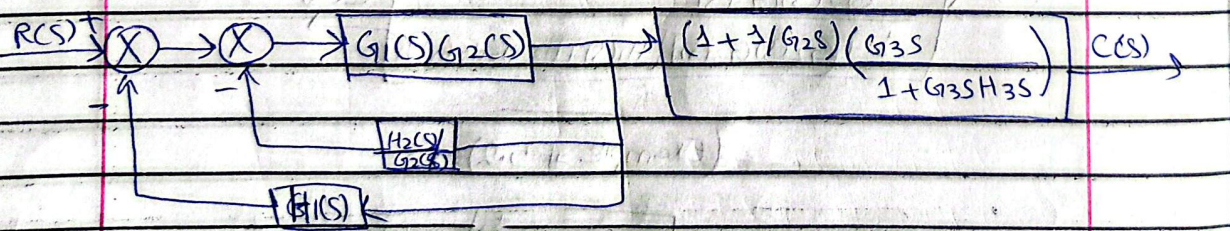


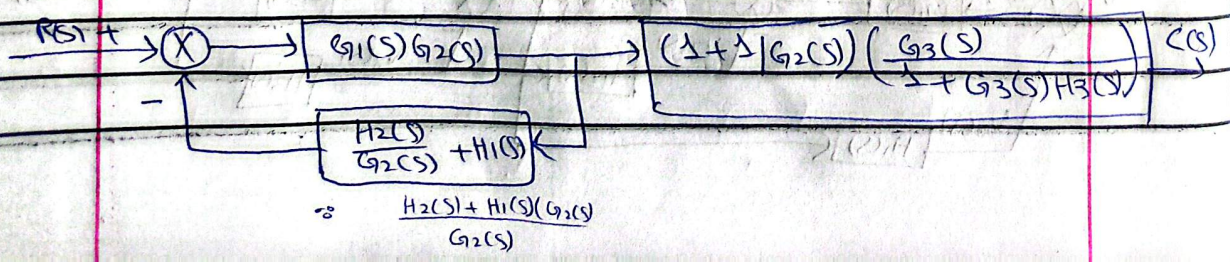
moving pick point before clock.



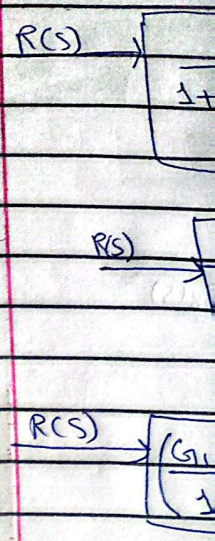




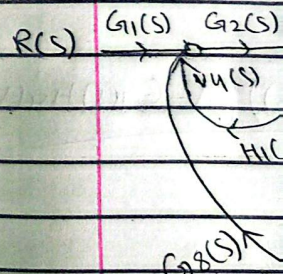




$\frac{H_2(S) + H_1(S)(G_2(S)}{G_2(S)}$



find the graph



$R(s) \rightarrow \dfrac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)\left[\dfrac{H_2(s)}{G_1(s)} + H_1(s)\right]} \rightarrow \dfrac{(1 + 1/G_2(s))/\left(\dfrac{G_3(s)}{1+G_3(s)H_3(s)}\right)}{} \rightarrow C(s)$

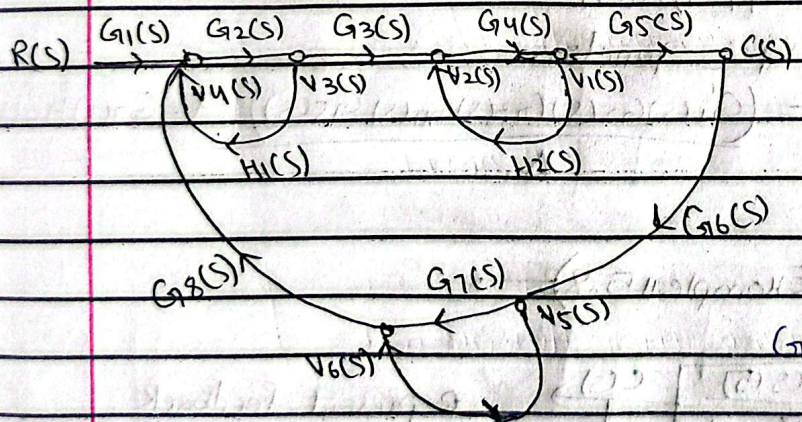
$R(s) \rightarrow \dfrac{G_1(s)G_2(s)}{1 + H_2(s)G_2(s) + H_1(s)G_1(s)G_2(s)} \rightarrow \dfrac{(1+1/G_2s)\left(\dfrac{G_3(s)}{1+G_3sH_3s}\right)}{} \rightarrow C(s)$

$R(s) \rightarrow \left(\dfrac{G_1(s)G_2(s)}{1+H_2sG_2s+H_1sG_1sG_2s}\right)\left(\dfrac{1+G_2s}{G_2s}\right)\left(\dfrac{G_3(s)}{1+G_3(s)H_3(s)}\right) \rightarrow C(s)$

$R(s) \rightarrow \dfrac{G_1(s)G_3(s)[1+G_2(s)]}{(1+H_2(s)G_2(s)+H_1(s)G_1(s)G_2(s))(1+G_3(s)H_3(s))} \rightarrow C(s)$

Example # 5.7B

Find the transfer function $C(s)/R(s)$, for the signal flow graph



Masons Rule formula.

$$G(s) = \frac{\sum P_i \Delta_i}{\Delta}$$

P_i = Gain of ith forward path.

Δ = determinant of the graph = 1 - (individual loop gains) + (Σ gain products of two non-touch loops) - (Σ gain product of three non-touching loops) + ...

Δ_i = determinant of the graph with loops that not touching i^{th} forward path removed.

- identify the forward path:

$P_1 = G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)$

- individual loop gains:

Loop 1: $G_2(s)H_1(s)$

Loop 2: $G_4(s)H_2(s)$

Loop 3: $G_7(s)H_4(s)$

Loop 4: $G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)$

- non-touching loops

loop 1 & loop 2: $G_2(s)H_1(s)G_4(s)H_2(s)$.

loop 1 & loop 3: $G_2(s)H_1(s)G_7(s)H_4(s)$

loop 2 & loop 3: $G_4(s)H_2(s)G_7(s)H_4(s)$

Three loops at a time:

loop 1 loop 2 loop 3 = $G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)$.

System determinant (Δ)

$$\Delta = 1 - [G_2(s)H_1(s) + G_4(s)H_2(s) + G_7(s)H_4(s) + (G_2(s)G_3(s)G_4G_5G_6G_7G_8)] + [G_2(s)H_1(s)G_4(s)H_2(s) + G_2(s)H_1(s)G_7(s)H_4(s) + G_4(s)H_2(s)G_7(s)H_4(s)] - [G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)]$$

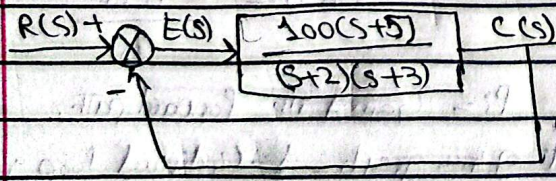
forward path determinant

$\Delta_1 = 1 - G_7(s)H_4(s)$

final transfer function:

$$G(s) = \frac{P_1\Delta_1}{\Delta} = \frac{(G_1(s)G_2(s)G_3(s)G_4(s)G_5(s))(1 - G_7(s)H_4(s))}{\Delta}$$

Example#5.8B



Represent feedback control system in state space. Model the forward transfer function in cascade form.

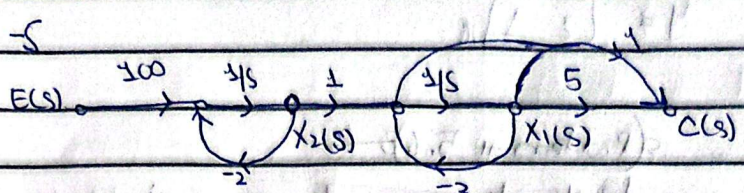
$$G_1(s) = \frac{100(s+5)}{(s+2)(s+3)}$$

step 1: Model the forward transfer function

denominator $(s+2)(s+3)$ gives poles at $s=-2$ and -3

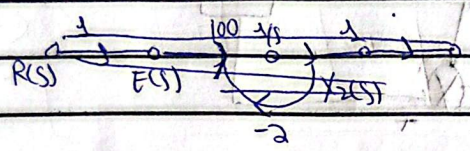
and overall multiplier provides a gain of 100.

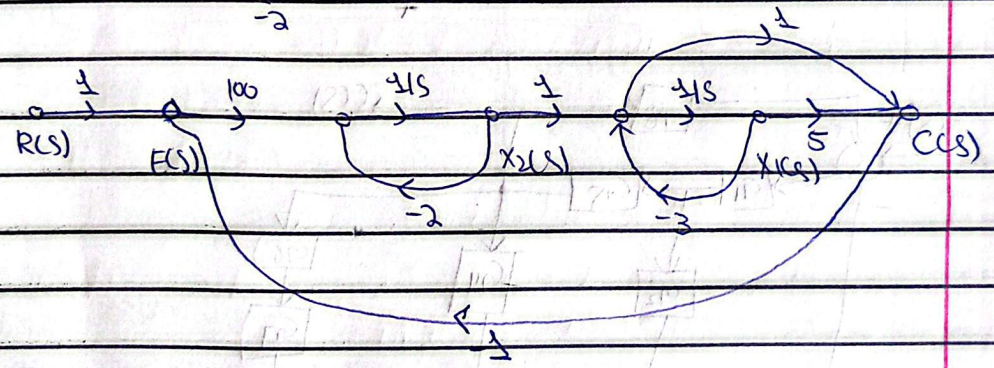
Zeros: The numerator term $(s+5)$ introduces a zero at -5



step 2: Add the feedback and i/p paths.

To convert the open loop path into a closed-loop feedback system, we add an i/p R(s) and o/p C(s) back to the i/p summing junction with a feedback gain of -1.





step 3: initial state equations:

$$\dot{x}_1 = -3x_1 + x_2$$

$$\dot{x}_2 = -2x_2 + 100(r - c)$$

step 4: determine output equation.

$$c = 5x_1 + (x_2 - 3x_1) = \boxed{2x_1 + x_2}$$

Substituting output into state equation.

$$\dot{x}_2 = -2x_2 + 100(r - 2x_1 - x_2)$$

$$\dot{x}_2 = -200x_1 - 102x_2 + 100r$$

final vector matrix form

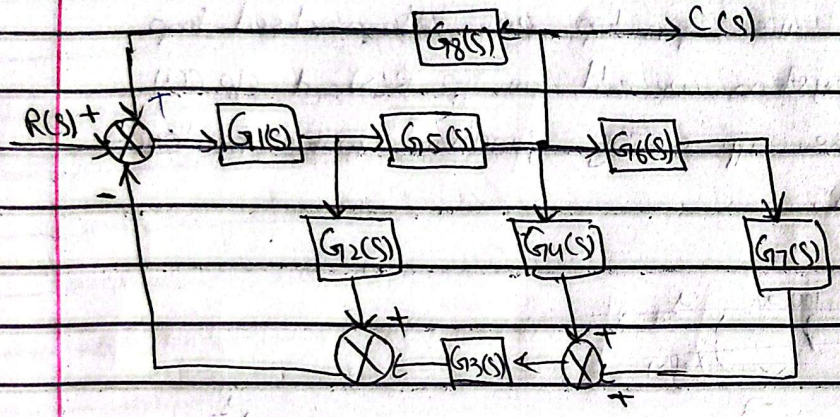
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -200 & -102 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 100 \end{bmatrix} u$$

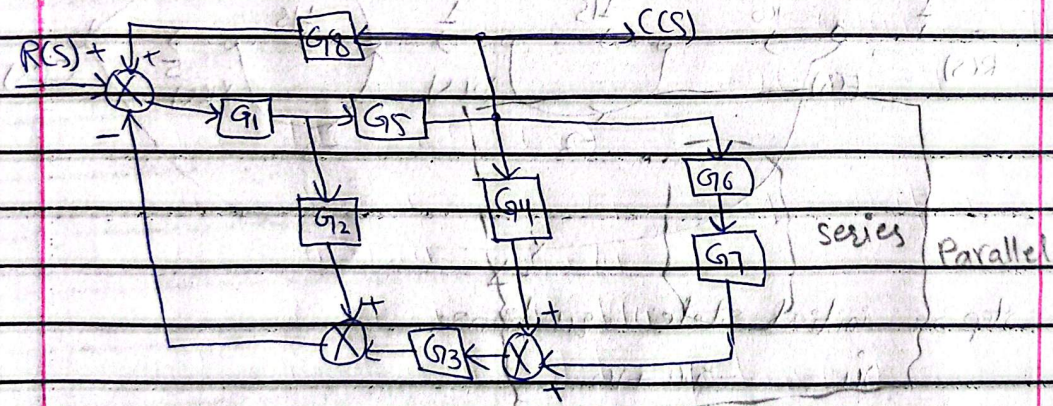
$$y = \begin{bmatrix} 2 & 1 \end{bmatrix} x.$$

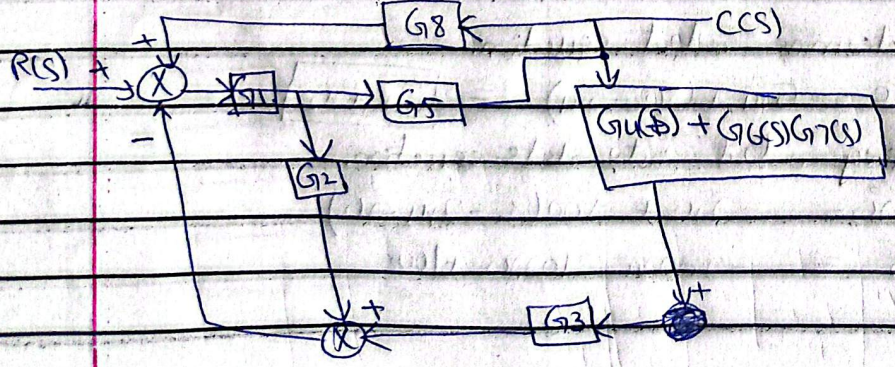
Problem # 5.6B

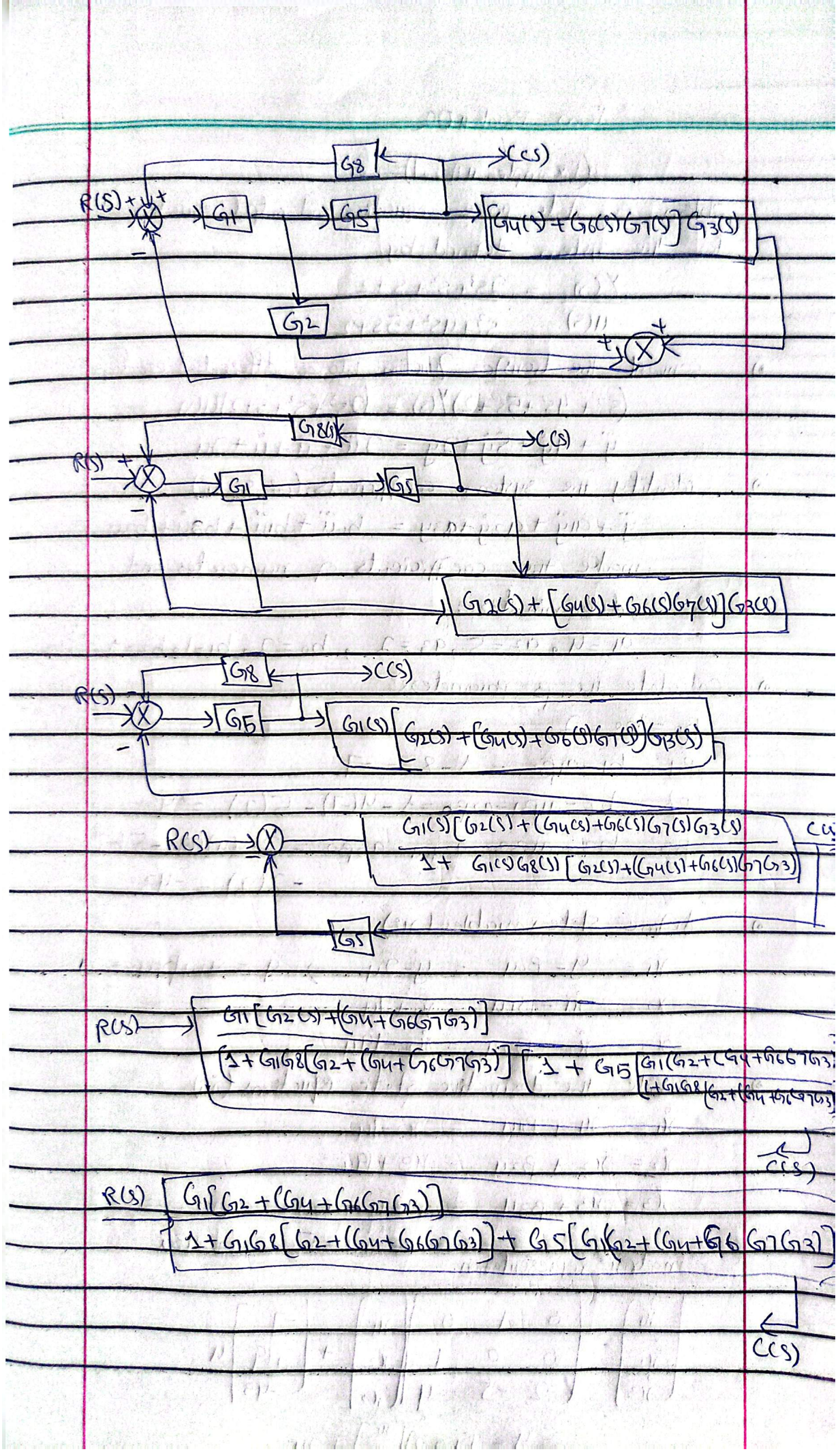
Reduce the block diagram to a single block

$T(s) = C(s)/R(s)$








R(s)
G1
G2
G5
G8
C(s)
[G4(s) + G6(s)G7(s)] G3(s)
R(s)
G1
G5
G8
C(s)
G2(s) + [G4(s) + G6(s)G7(s)]G3(s)
R(s)
G5
G8
C(s)
G1(s)[G2(s) + (G4(s) + G6(s)G7(s)G3(s)
R(s)
G5
C(s)
R(s)
G1[G2(s) + (G4 + G6G7G3)]
C(s)
R(s)
G1[G2 + (G4 + G6G7G3)]
C(s)

from Book #02

(Example #A2.7)B

Obtain a state space equation and output equation for the system defined by

$$\frac{Y(s)}{U(s)} = \frac{2s^3+s^2+s+2}{s^3+4s^2+5s+2}$$

a) convert the transfer function into a differential equation.

$$(s^3+4s^2+5s+2)Y(s) = (2s^3+s^2+s+2)U(s)$$

$$\dddot{y}+4\ddot{y}+5\dot{y}+2y = 2\ddot{u}+\dot{u}+\dot{u}+2u$$

a) identify the system coefficients (a_i & b_i)

$$\dddot{y}+a_1\ddot{y}+a_2\dot{y}+a_3y = b_0\ddot{u}+b_1\dot{u}+b_2\dot{u}+b_3u$$

make the coefficients of numerator and denominator:

$a_1=4,\ a_2=5,\ a_3=2,\ b_0=2,\ b_1=1,\ b_2=1,\ b_3=2$

a) Calculate the β parameters

$\beta_0 = b_0 = 2$

$\beta_1 = b_1 - a_1\beta_0 = 1-8 = -7$

$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 = 1-4(-7)-5(2) = 19$

$\beta_3 = b_3 - a_1\beta_2 - a_2\beta_1 - a_3\beta_0 = 2-(4)(19)-5(-7)-2(2) = -43$

a) define state variables (x_i)

$x_1 = y-\beta_0 u = y-2u \Rightarrow y = x_1+\beta_0 u = x_1+2u$

$x_2 = \dot{x}_1 - \beta_1 u = \dot{x}_1+7u$

$x_3 = \dot{x}_2 - \beta_2 u = \dot{x}_2 - 19u$

a) write the derivative state equations ($\dot{x}_i$)

$\dot{x}_1 = x_1+\beta_1 u = x_2-7u$

$\dot{x}_2 = x_2+\beta_2 u = x_3+19u$

$\dot{x}_3 = x_3+\beta_3 u = -a_3x_1-a_2x_2-a_1x_3+\beta_3 u$

$= -2x_1-5x_2-4x_3-43u$

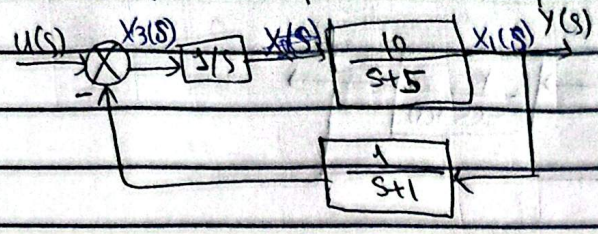
matrix representation.

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -2 & -5 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}-7\\ +19\\ -43\end{bmatrix}u$$

$$y = [1\ 0\ 0]\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + 2u$$

Example A-2-8B

obtain a state space model of the system.



x_1: $\frac{X_1(S)}{X_2(S)} = \frac{10}{S+5}$

x_2: $\frac{X_2(S)}{U(S)-X_3(S)} = \frac{1}{S}$

x_3: $\frac{X_3(S)}{X_1(S)} = \frac{1}{S+1}$

y: $Y(S) = X_1(S)$

write equations in frequency domain

a) x_1: $\frac{X_1(S)}{X_2(S)} = \frac{10}{S+5}$

$10 X_2(S) = (S+5) X_1(S)$

$S X_1(S) = 10 X_2(S) - 5 X_1(S)$

b) x_2: $X_2(S) S = U(S) - X_3(S)$

$S X_2(S) = -X_3(S) + U(S)$

c) x_3: $(S+1) X_3(S) = X_1(S)$

$S X_3(S) + X_3(S) = X_1(S)$

$S X_3(S) = X_1(S) - X_3(S)$

d) y: $Y(S) = X_1(S)$

convert to the time domain

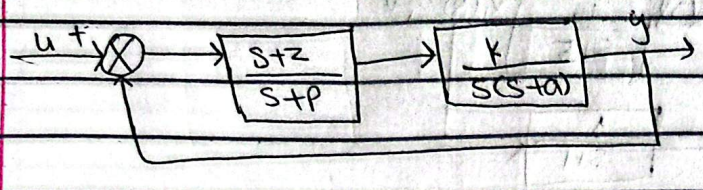
$y = x_1$, $\dot{x}_1 = -5x_1 + 10x_2$, $\dot{x}_2 = -x_3 + u$, $\dot{x}_3 = x_1 - x_3$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -5 & 10 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

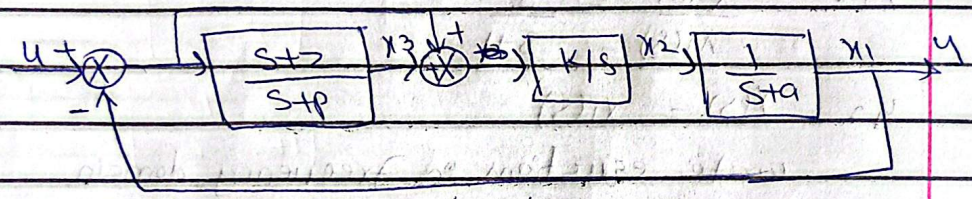
Example (A-2-10)

obtain a state-space representation of the system



$$\frac{S+z}{S+p} = 1 + \frac{z-p}{S+p}$$

$\frac{K}{S(S+a)} \rightarrow \frac{K}{S}$ → an integrator, $\frac{1}{S+a}$ → first order pole



$X_1(S)/X_2(S) = 1/S+a$, $X_2(S)/X_3(S) = K/S$, $X_3(S)/U(S) = \frac{S+z}{S+p}$

state variable

$y = x_1 = \frac{1}{S+a}$, $x_2 = \frac{K}{S}$, $x_3 = \frac{S+z}{S+p}$

$X_2(S) = SX_1(S) + aX_1(S) \Rightarrow X_2(S) - aX_1(S)$

$\dot{x}_1 = -ax_1 + x_2$

$\dot{x}_2 = -Kx_1 + Kx_3 + Ku$

$\dot{x}_3 = -(z-p)x_1 - px_3 + (z-p)u$

$y = x_1$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -a & 1 & 0 \\ -K & 0 & K \\ -(z-p) & 0 & -p \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ K \\ z-p \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(Example A-2-11)B

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

obtain the transfer function system.

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, D = 0$$

$$G(s) = C(sI - A)^{-1}B + D$$

$$= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} s+1 & -1 & 0 \\ 0 & s+1 & -1 \\ 0 & 0 & s+2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

let $M = \begin{bmatrix} s+1 & -1 & 0 \\ 0 & s+1 & -1 \\ 0 & 0 & s+2 \end{bmatrix}$

$$M^{-1} = \frac{\text{adj}(M)}{\det(M)}$$

$\det(M) = (s+1)(s+1)(s+2) = (s+1)^2(s+2)$

we get adjacent of M by Transpose the cofactor matrix

Row 1, 1st: $\begin{bmatrix} s+1 & -1 \\ 0 & s+2 \end{bmatrix} = (s+1)(s+2)$

2nd $= \begin{bmatrix} 0 & -1 \\ 0 & s+2 \end{bmatrix} = 0$

3rd $= \begin{bmatrix} 0 & s+1 \\ 0 & 0 \end{bmatrix} = 0$

1st

row 2: $\begin{bmatrix} -1 & 0 \\ 0 & s+2 \end{bmatrix} = +(s+2)$

2nd: $\begin{bmatrix} s+1 & 0 \\ 0 & s+2 \end{bmatrix} = (s+1)(s+2)$

3rd: $\begin{bmatrix} s+1 & -1 \\ 0 & 0 \end{bmatrix} = 0$

row 3:

1st $\rightarrow \begin{bmatrix} -1 & 0 \\ s+1 & -1 \end{bmatrix} = 1$

2nd $\rightarrow \begin{bmatrix} s+1 & 0 \\ 0 & -1 \end{bmatrix} = 0 + (s+1)$

3rd $\rightarrow \begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix} = 0 + (s+1)^2$

$$adj(M) = \begin{bmatrix} (s+1)(s+2) & s+2 & 1 \\ 0 & (s+1)(s+2) & s+1 \\ 0 & 0 & (s+1)^2 \end{bmatrix}$$

$$M^{-1} = \begin{bmatrix} 1/s+1 & 1/(s+1)^2 & 1/(s+1)^2(s+2) \\ 0 & 1/(s+1) & 1/(s+1)(s+2) \\ 0 & 0 & 1/(s+2) \end{bmatrix}$$

$$G(s) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/(s+1) & 1/(s+1)^2 & 1/(s+1)^2(s+2) \\ 0 & 1/(s+1) & 1/(s+1)(s+2) \\ 0 & 0 & 1/(s+2) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

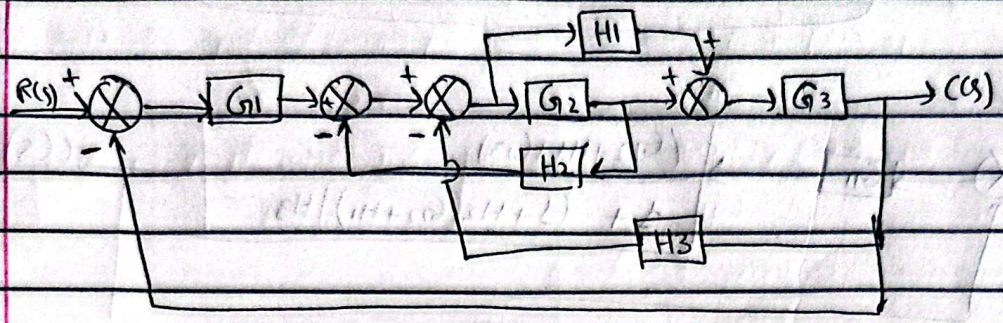
$$= \begin{bmatrix} 1/(s+1) & 1/(s+1)^2 & 1/(s+1)^2(s+2) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

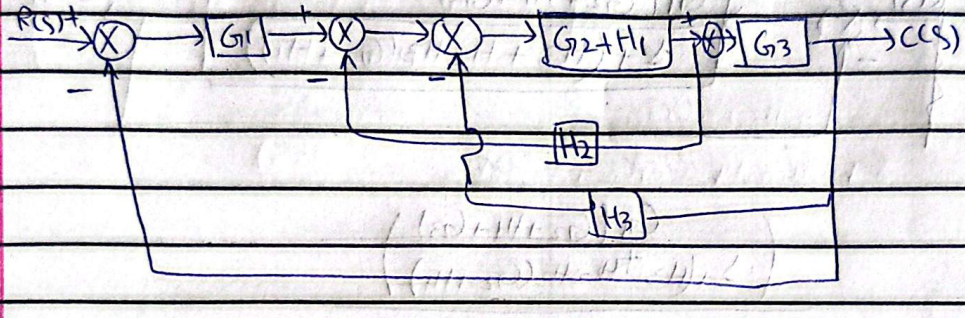
$$G(s) = \begin{bmatrix} 1/(s+1)^2(s+2) \end{bmatrix}$$

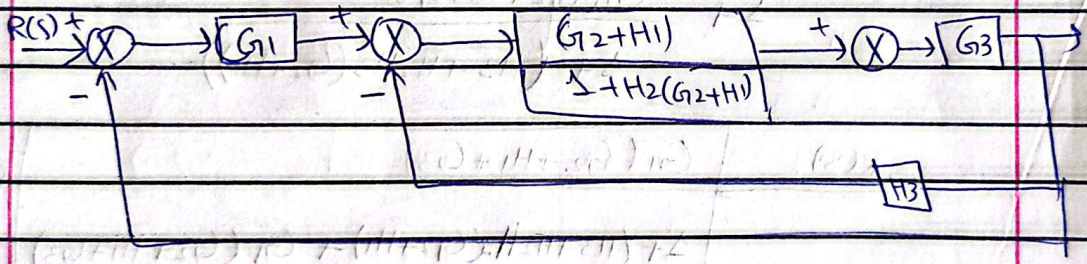
$$G(s) = \frac{1}{s^3 + 4s^2 + 5s + 2}$$

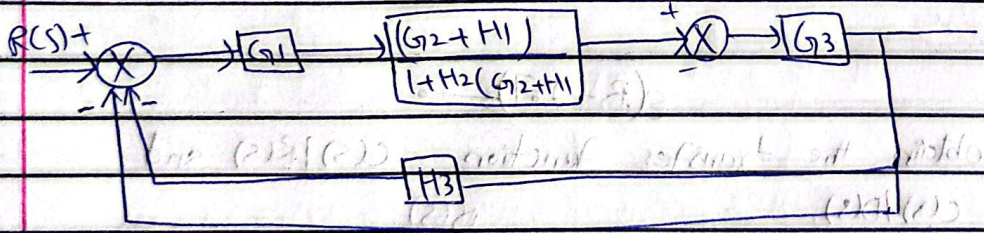
Exercise Problem : B-2.3

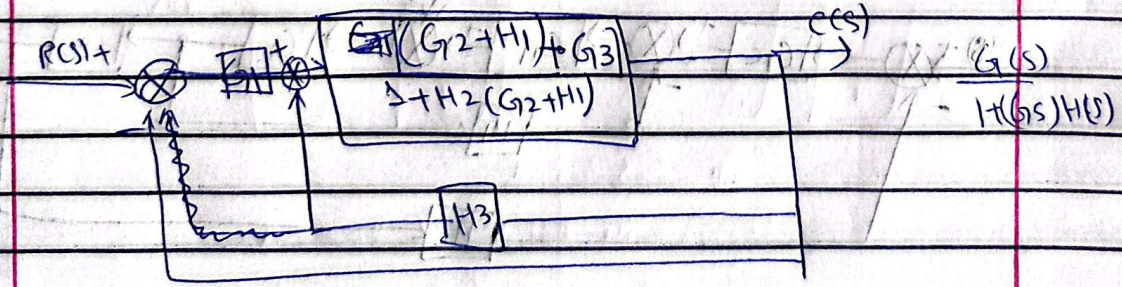
Simplify block diagram and obtain the closed loop transfer function C(s)/R(s)

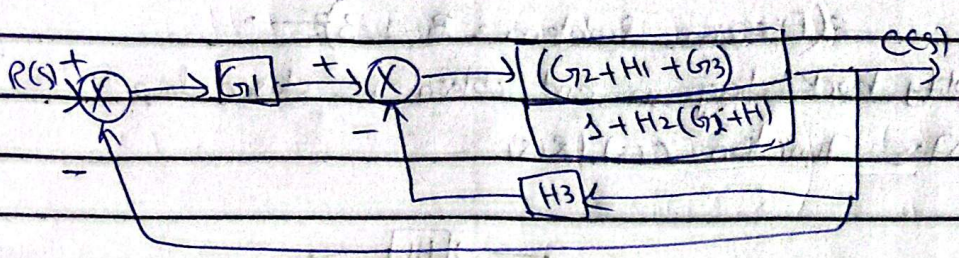


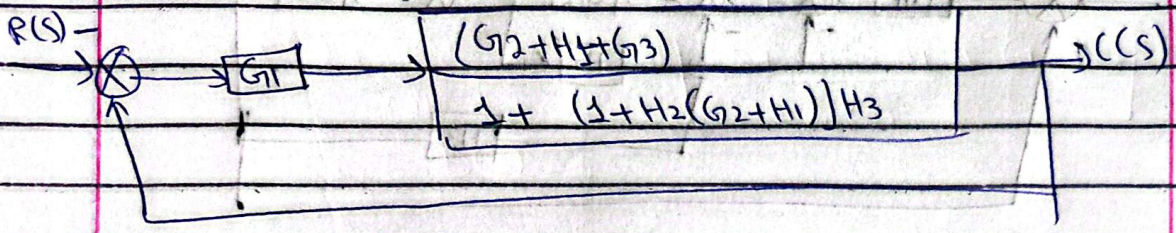


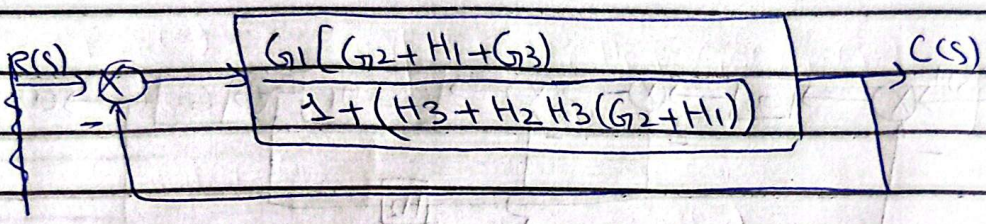




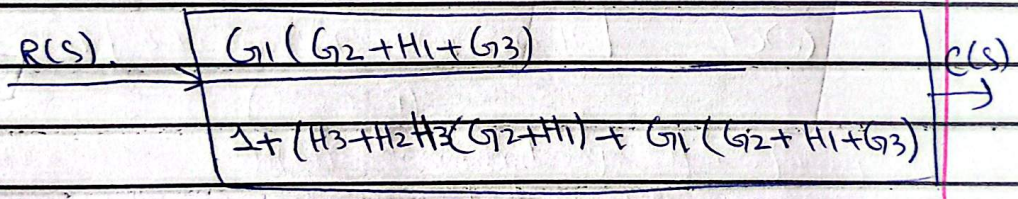




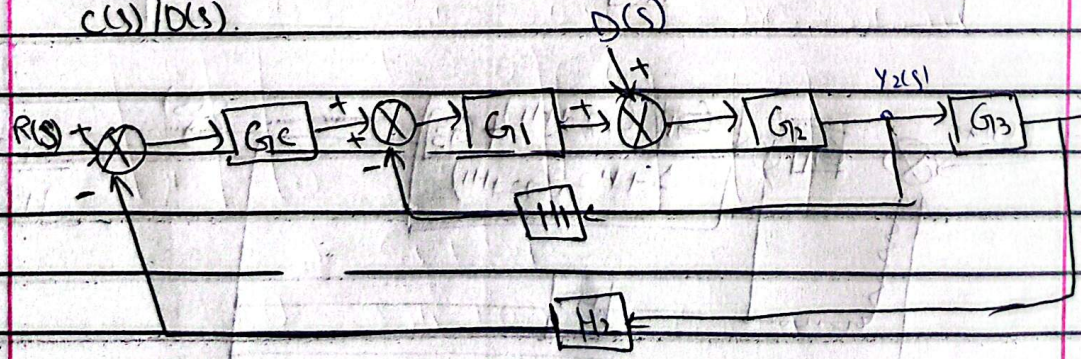




$$\left(\frac{G_1[G_2+H_1+G_3)}{1+(H_3+H_2H_3(G_2+H_1))}\right) \Big/ \left(1+\frac{G_1(G_2+H_1+G_3)}{1+(H_3+H_2H_3(G_2+H_1))}\right)$$



obtain the transfer function C(s)/R(s) and C(s)/D(s).



Error at 1st summing junction.

$E_1(s) = R(s) - H_2C(s)$ —(1)

output of $G_1(s)$:

$X_1(s) = G_1(s)E_1(s) = G_1(s)[R(s) - H_2C(s)]$ —(2)

output of second summing junction: (i/p to G_1)

$X_2(s) = X_1(s) - H_1Y_2(s)$ —(3)

output of 3rd summing junction (combine o/p of G_1 and $D(s)$)

$X_3(s) = G_1X_2(s) + D(s)$. —(4)

o/p of block $G_2(Y_2)$:

$Y_2(s) = G_2X_3(s) = G_2[G_1X_2(s) + D(s)]$ —(5)

final output $C(s)$:

$C(s) = G_3Y_2(s) = G_3G_2[G_1X_2(s) + D(s)]$ —(6)

$C(s) = G_1G_2G_3X_2(s) + G_2G_3D(s)$

$\boxed{\frac{C(s) - G_2G_3D(s)}{G_1G_2G_3} = X_2(s)}$ put in (5)

$Y_2(s) = G_2[G_1X_2(s) + D(s)]$

$Y_2(s) = G_1G_2X_2(s) + G_2D(s)$

$= G_1G_2\left[\frac{C(s) - G_2G_3D(s)}{G_1G_2G_3}\right] + G_2D(s)$

$Y_2(s) = \frac{C(s)}{G_3} - G_2D(s) + G_2D(s)$

$\boxed{Y_2(s) = \frac{C(s)}{G_3}}$ substitute $Y_2(s)$ and $X_2(s)$ in (3)

$X_1(s) = X_2(s) + H_1Y_2(s)$

$= \frac{C(s) - G_2G_3D(s)}{G_1G_2G_3} + H_1\frac{C(s)}{G_3}$

$X_1(s) = \frac{C(s)}{G_1G_2G_3} + H_1\frac{C(s)}{G_3} - \frac{G_2G_3D(s)}{G_1G_2G_3}$

$X_1(s) = C(s)\left[\frac{1 + G_1G_2H_1}{G_1G_2G_3}\right] - \frac{D(s)}{G_1}$ —(7)

substitute $X_1(s) = G_c[R(s) - H_2 C(s)]$ in (1)

$$G_c R(s) - G_c H_2 C(s) = Cs\left(\frac{1 + G_1G_2H_1}{G_1G_2G_3}\right) - \frac{D(s)}{G_1}$$

$$G_c R(s) + \frac{D(s)}{G_1} = Cs\left(\frac{1 + G_1G_2H_1}{G_1G_2G_3}\right) + G_c H_2 C(s)$$

$$G_c R(s) + \frac{D(s)}{G_1} = C(s)\left[\frac{1 + G_1G_2H_1}{G_1G_2G_3} + G_c H_2\right]$$

assume $D(s) = 0$

$$\frac{G_c R(s)}{1} = C(s)\left[\frac{1 + G_1G_2H_1 + G_cG_1G_2G_3H_2}{G_1G_2G_3}\right]$$

$$\frac{G_cG_1G_2G_3}{1 + G_1G_2H_1 + G_cG_1G_2G_3H_2} = \frac{C(s)}{R(s)}$$

assume $R(s) = 0$

$$\frac{D(s)}{G_1} = C(s)\left[\frac{1 + G_1G_2H_1 + G_cG_1G_2G_3H_2}{G_1G_2G_3}\right]$$

$$\frac{G_1G_2G_3}{G_1\left[1 + G_1G_2H_1 + G_cG_1G_2G_3H_2\right]} = \frac{C(s)}{D(s)}$$

consider the system described by $\dddot{y} + 3\ddot{y} + 2\dot{y} = u$ drive a state space representation of the system.

$x_1 = y$, $x_2 = \dot{x}_1 = \dot{y}$, $x_3 = \dot{x}_2 = \ddot{y}$, $x_4 = \dot{x}_3 = \dddot{y}$

$$\dddot{y} = -2\dot{y} - 3\ddot{y} + u$$

$$\dot{x}_3 = -2\dot{x}_1 - 3\dot{x}_2 + u = -2x_2 - 3x_3 + u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = Cx + Du$$

$$y(t) = 1 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 0 \cdot u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + 0(u)$$

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